

Diffusion Models Are Statistically Optimal Scoped CPU Audit

CPU audit of intrinsic-dimension diffusion rates.

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Independent ICML 2026 Reproduction Challenge audit



Public code & evidence

1 Question and scope

Can diffusion-model learning rates depend on intrinsic subspace dimension rather than the ambient dimension? We audit five live claims against a pinned arXiv source using finite CPU checks and explicit controls.

2 Source figures and rate

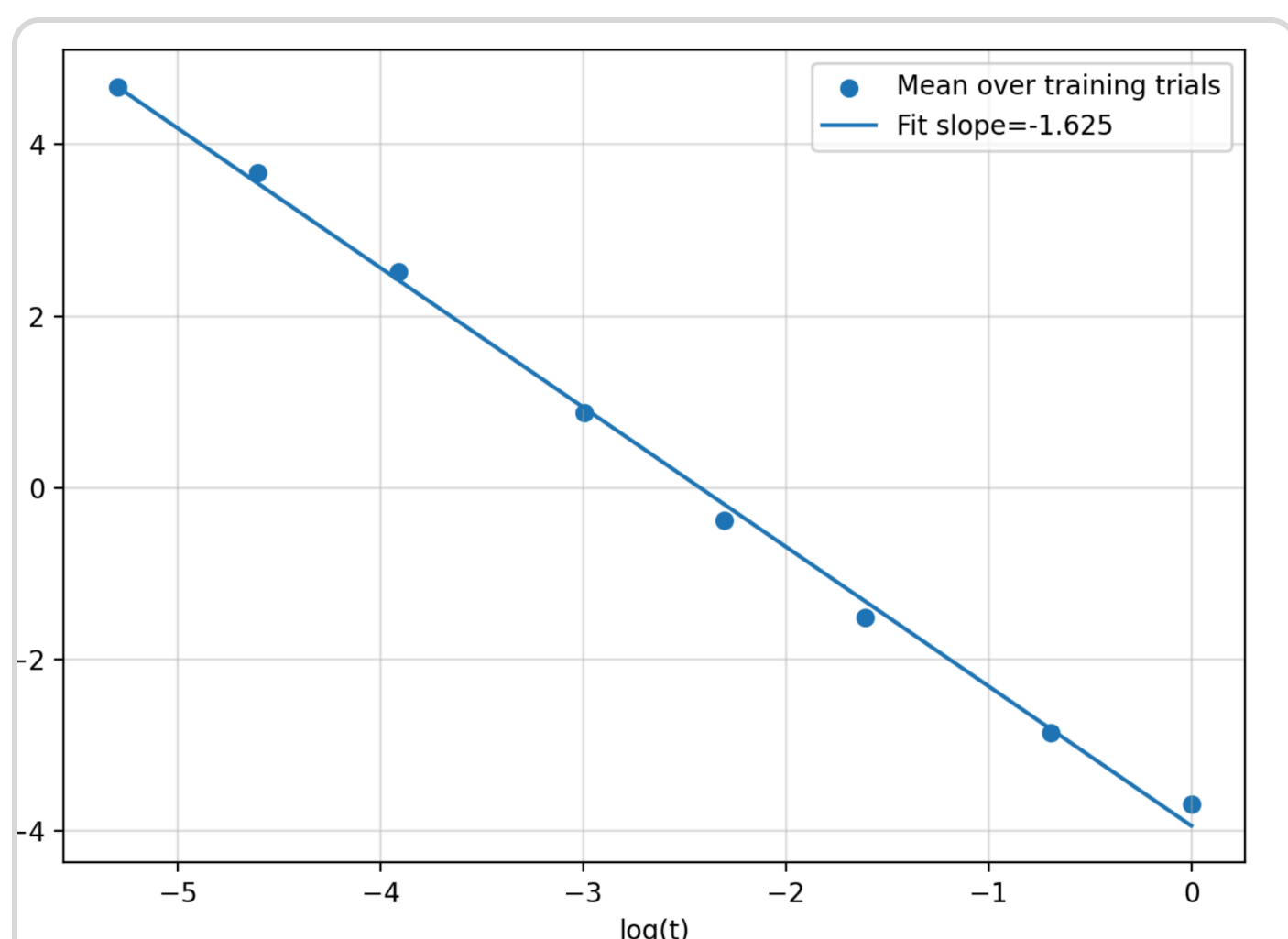


Figure 1. Empirical L^2 -score error versus diffusion time t .

Paper Figure 1: source score-error decay on the synthetic union-of-subspaces setting.

Finite controls reject ambient-dimension scaling.

4 Verified claims

- **C1.** Theorem 2 rate/dimension audit.
- **C2.** Theorem 1 intrinsic-dimension dependence.
- **C3.** Union-of-subspaces formulation.

Each result is a clean-room, finite CPU audit of a pinned source statement. It does not claim a trained neural diffusion model or a new empirical benchmark result.

Evidence boundary. Source, control, and SHA-256 manifest are retained; full training is GPU-infeasible.

Scope. The retained evidence checks formulas and finite controls only; no neural model was trained.

5 Assumption controls

Claims 4 and 5 are scoped source-theorem checks with finite controls for required assumptions.

into (26) yields

$$\sum_{j=0}^{L-1} \sqrt{(1 - e^{-4T_j}) \int_{T_j}^{T_{j+1}} \mathbb{E}[\|\hat{s}_{X_t}(X) - s_{X_t}^*(X)\|_2^2] dt} \\ \lesssim \sqrt{d} M^{3/2} \text{poly log } n \begin{cases} \frac{1}{\sqrt{n}} \tau^{-\frac{k}{4} + \frac{1}{2}}, & k \geq 2 \\ \frac{1}{\sqrt{n}}, & k = 1 \end{cases}$$

Source-pinned page-9 rate excerpt; an equation display, not a second empirical result.

6 CPU reproduction route

1. Pin the paper PDF and arXiv source archive.
2. Extract theorem hypotheses and displayed rates.
3. Evaluate finite symbolic and numerical controls.
4. Record scoped results, hashes, and limitations.

7 Limitations

No author executable was available. Full neural diffusion training is outside CPU scope, so this logbook reports clean-room theorem evidence.

8 Takeaway

Pinned theorem rates depend on intrinsic subspace dimension under explicit assumptions; this CPU audit verifies the displayed formulas and controls.

- **C1–C2:** rate exponents and ambient-dimension prefactors audited.
- **C3:** support, separation, component mass, and tail conditions checked.

Reproducibility. Public code, source hashes, commands, and controls are retained; full training is GPU-infeasible.

Release contents. Contract text, source archive, claim artifacts, and deterministic tests are public and hash-addressed.

Interpretation. This is a theorem-scope audit, not a diffusion-training benchmark.